(b) • The length of wire W between P and Q is L.

Measure and record L.

L = ...

• Close the switch.

• The potential difference V across the LED is given by the voltmeter.

The resistance R of the LDR is given by the ohmmeter.

Measure and record V and R.

V = ...

R = ...

• Open the switch.

[2]

(c) • Change the length of wire W between P and Q so that L is approximately 90 cm.

• Repeat **(b)**.

L = ...

V = ...

R = ...

[3]

(d) It is suggested that the relationship between V, d and R is

$$V = \frac{Zd}{R} + k$$

where Z has the value $1.00 \times 10^{3}\,\mathrm{V\,\Omega\,m^{-1}}$ and k is a constant.

(i) Using your data, calculate **two** values of k.

first value of k = ...

second value of k = ...

[1]